CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-501: Introduction to DBMS

Session: December 2018 Examination | **Paper Code:** 18021 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) **Define Super Keys, Candidate Keys, and Primary Keys.**

Definitions:

- **Super Key:** A set of one or more attributes that uniquely identifies each tuple in a relation (may contain redundant attributes).
- **Candidate Key:** A minimal super key having no extraneous/redundant attributes.
- **Primary Key:** The specific candidate key chosen by the database designer as the principal tuple identifier. Cannot contain `NULL`.

Q2 (3M) **Explain the term "Data Independence".**

Definition: Data Independence is the capacity to change the schema at one level of the database system without having to alter the schema at the next higher level. It includes Physical Data Independence (modifying internal storage without affecting conceptual schema) and Logical Data Independence (modifying conceptual schema without altering external views).

Q3 (3M) **What is Hashing? Explain.**

Definition: Hashing is a direct addressing technique where a mathematical algorithm (hash function) calculates the exact physical block address of a record from its search key value, enabling O(1) constant-time retrieval.

Q4 (3M) **What do you mean by Relational Database? Describe.**

Definition: A **Relational Database** is a collection of data items organized into formal tables (relations) of rows and columns based on first-order predicate logic, where relations are linked together via primary and foreign key constraints.

Q5 (3M) **Briefly explain the terms Security and Authorization.**

Definitions:

- **Database Security:** Protection of the database against unauthorized access, malicious attacks, corruption, or destruction.
- **Authorization:** The mechanism that defines and enforces what actions (e.g., SELECT, INSERT, UPDATE, DROP) a validated user or role is permitted to perform on specific database objects.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) **What is a Referential Integrity Constraint? Why do we need it? Explain with a suitable example.**

Explanation: Referential integrity requires that a foreign key value in a referencing table must match an existing primary key value in the referenced table, or be `NULL`.
Need: Prevents orphan records and inconsistent states.
Example:

```
-- Parent Table
CREATE TABLE Department (DeptNo INT PRIMARY KEY, DName VARCHAR(30));

-- Child Table with Foreign Key
CREATE TABLE Employee (
    EmpID INT PRIMARY KEY,
    EmpName VARCHAR(50),
    DeptNo INT,
    CONSTRAINT fk_emp_dept FOREIGN KEY (DeptNo) REFERENCES Department(DeptNo)
        ON DELETE CASCADE ON UPDATE CASCADE
);
```

If department 10 is deleted, `ON DELETE CASCADE` automatically removes all employees assigned to department 10, preventing dangling references.

Q7 (7.5M) **List four significant differences between a File-Processing System and a DBMS.**

Differences:

Aspect	File-Processing System	DBMS
Data Redundancy	High redundancy across duplicate files.	Controlled and normalized.
Data Independence	None (tight coupling with code).	Logical & Physical Data Independence.
Concurrency	Prone to race conditions and lost updates.	ACID compliant concurrency control.
Crash Recovery	Manual restore; inconsistent state.	Automated log-based rollback and redo.

Q8 (7.5M) **Explain Third Normal Form (3NF) with example.**

Definition: A relation is in **3NF** if it is in 2NF and has no transitive functional dependencies for non-prime attributes. Formally, for every non-trivial FD X \to Y, either X is a super key or Y is a prime attribute.
Example: `Student(RollNo, Name, DeptID, DeptHead)`.
RollNo \to DeptID and DeptID \to DeptHead (Transitive!).
Decompose into: `Student(RollNo, Name, DeptID)` and `Department(DeptID, DeptHead)`.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **Explain the different locking techniques with examples for concurrency control.**

Locking Techniques:

1. **Shared Lock (S-Lock / Read Lock):** Multiple transactions can simultaneously hold shared locks to read data without conflict.
2. **Exclusive Lock (X-Lock / Write Lock):** Only one transaction can hold an exclusive lock to read and write. No other transaction can hold any lock on that item.
3. **Two-Phase Locking (2PL):** Growing Phase (acquiring locks) followed by Shrinking Phase (releasing locks). Guarantees conflict serializability.
4. **Strict 2PL:** Exclusive locks held until transaction commits, eliminating cascading rollbacks.

Q10 (15M) **Explain Indexed Sequential, B-Tree, and B+ Tree file organizations with examples.**

Comparison of File Organizations:

- **Indexed Sequential (ISAM):** Records stored in sequential key order with a multilevel static index. Good for sequential and direct access, but degrades with many overflow records.
- **B-Tree:** Self-balancing dynamic multi-way search tree. Keys and data record pointers reside in both internal nodes and leaf nodes.
- **B+ Tree:** Modern standard for relational DBMS storage engines. Internal nodes store only search keys to guide navigation, while **all actual data records/pointers reside in leaf nodes** connected via a doubly linked list, enabling superior range scans.

Q11 (15M) **Explain Super Class, Sub Class, Inheritance, and Generalization with examples.**

Enhanced ER (EER) Concepts:

- **Superclass:** A generic entity set that includes distinct subgroups (e.g., `PERSON`).
- **Subclass:** A specialized sub-entity set that inherits all attributes and relationships from its superclass, while adding specialized attributes (e.g., `STUDENT` with GPA, `TEACHER` with Rank).
- **Attribute Inheritance:** Subclass entities automatically inherit the primary key and all general attributes of their superclass.
- **Generalization:** Bottom-up synthesis where lower-level entity sets are combined into a generalized superclass entity.

Q12 (15M) **Write short notes on: (i) Strong and Weak Entity, (ii) Different Levels of Data Abstraction, (iii) Database Recovery Techniques.**

(i) Strong vs Weak Entity: A Strong entity has a primary key of its own. A Weak entity does not have a primary key and depends on the existence of an identifying strong entity set (identified by discriminator + strong entity PK).
(ii) Data Abstraction Levels: Physical level (low-level block storage), Logical level (community tables and relations), and View level (user-specific abstractions).
(iii) Database Recovery Techniques: Log-based recovery (Write-Ahead Logging), Deferred/Immediate update schemes, and Checkpoints to restore consistency following system failures.

Q13 (15M) **A hospital maintains data about entities: PATIENTS, DOCTORS, BEDS, ACCOUNTS. Construct an E-R diagram specifying keys, mapping cardinalities, and participation constraints. Show table mapping.**

Entities & Attributes:

- `PATIENTS(SSNo, LastName, FirstName, HomePhone, Sex, DOB, Age, Street, City, State, Zip)`
- `DOCTORS(SSNo, LastName, FirstName, OfficePhone, Pager, Specialty)`
- `BEDS(RoomNumber, BedNumber, Type, Status, PricePerHour)`
- `ACCOUNTS(AccountNo, DateIn, DateOut, TotalAmount, Status)`

Relationships:

- `TREATS` (M:N between DOCTORS and PATIENTS)
- `ASSIGNED_BED` (1:1 between PATIENTS and BEDS)
- `HAS_ACCOUNT` (1:1 between PATIENTS and ACCOUNTS)

